

RC low-pass filter: frequency domain

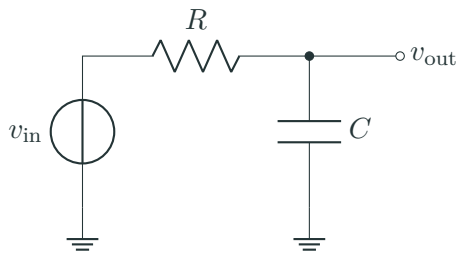
A resistor and capacitor smooth a changing input. Take the output across the capacitor: slow changes pass through, while fast changes are attenuated.

Assumption Ideal voltage source, ideal components, and no output load. For the step response, the capacitor is initially uncharged.

Transfer function and cutoff

$$H(s) = \frac{V_{\text{out}}(s)}{V_{\text{in}}(s)} = \frac{1}{1 + sRC}, \quad \tau = RC, \quad f_c = \frac{1}{2\pi RC}$$

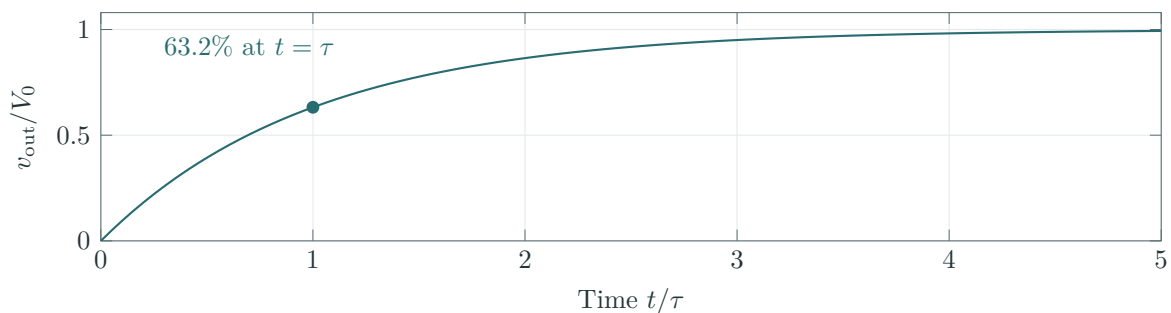
Quantity	Meaning	Value
R	Series resistance	1.0 k Ω
C	Shunt capacitance	100 nF
τ	Time constant	100 μ s
f_c	Half-power cutoff frequency	1.59 kHz



Step response: time domain

Response to a step of amplitude V_0

$$v_{\text{out}}(t) = V_0(1 - e^{-t/\tau}), \quad t \geq 0$$



Problem-set check. At DC, the gain tends to one. At the cutoff, $|H| = 1/\sqrt{2}$ and the phase is -45° . Use $\omega_c = 1/RC$ for rad/s and $f_c = 1/(2\pi RC)$ for Hz.